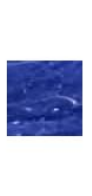
```
Arguments used:
  folder: C:\Users\pensf\Documents\AutomatedLBP_Benchmarking\LBP_Test_Images\Swatches\LBP_Texture_Swatches
  P: 8
  R: 1.0
  method: uniform
  metric: cosine
  save_csv: matches.csv
  blur: 1.0
  ltp: False
  visualize: True
  X: 64
  Y: 64
  V: True
  G: False
Correct matches: 28/75 (37.33%)
Highest distance among correct matches: 0.005695
Lowest distance among correct matches: 0.000099
Highest distance among incorrect matches: 0.002585
Lowest distance among incorrect matches: 0.000087
```

```
Index: 0
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000108
```



```
Index: 1
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000321
```

```
Index: 2
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000486
```

```
Index: 3
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000569
```

```
Index: 4
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000281
```

```
Index: 5
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000087
```

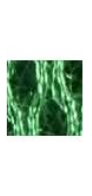
```
Index: 6
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.001274
```

```
Index: 7
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000683
```

```
Index: 8
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Cosine
Distance: 0.000157
```

```
Index: 9
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Cosine
Distance: 0.000157
```

```
Index: 10
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000581
```



```
Index: 11
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.000845
```

```
Index: 12
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.001437
```

```
Index: 13
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.002585
```

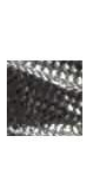
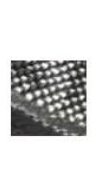
```
Index: 14
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.000346
```

```
Index: 15
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000351
```

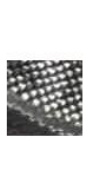
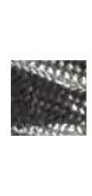
```
Index: 16
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.000396
```

```
Index: 17
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.000904
```

```
Index: 18
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.005695
```

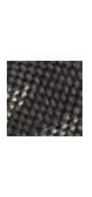



```
Index: 19
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.000980
```

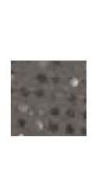



```
Index: 20
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.000980
```

```
Index: 21
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.001764
```



```
Index: 22
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.005695
```



```
Index: 23
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.000355
```

```
Index: 24
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.000814
```

```
Index: 25
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.000597
```

```
Index: 26
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000859
```

```
Index: 27
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.000773
```

```
Index: 28
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.001293
```

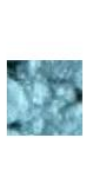
```
Index: 29
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000142
```

```
Index: 30
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.000167
```

```
Index: 31
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.000104
```

```
Index: 32
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.000221
```

```
Index: 33
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.000247
```



```
Index: 34
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Cosine
Distance: 0.000582
```

```
Index: 35
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.000230
```

```
Index: 36
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.000322
```

```
Index: 37
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.000334
```

```
Index: 38
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.000167
```

```
Index: 39
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.000312
```

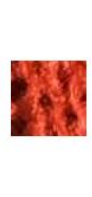
```
Index: 40
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.000230
```

```
Index: 41
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000153
```

```
Index: 42
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000262
```

```
Index: 43
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000281
```

```
Index: 44
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000321
```

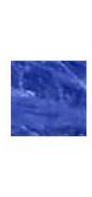


```
Index: 45
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000411
```

```
Index: 46
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000244
```

```
Index: 47
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.000244
```

```
Index: 48
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.000247
```

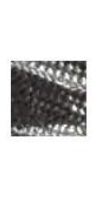


```
Index: 49
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.000087
```

```
Index: 50
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000253
```

```
Index: 51
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000452
```

```
Index: 52
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.000362
```



```
Index: 53
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.001446
```

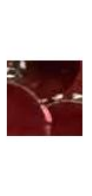
```
Index: 54
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.000785
```

```
Index: 55
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.000437
```

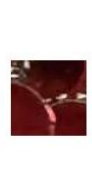
```
Index: 56
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.000437
```

```
Index: 57
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.000773
```

```
Index: 58
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.000494
```



```
Index: 59
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000328
```

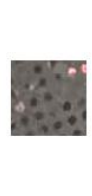


```
Index: 60
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000328
```

```
Index: 61
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.002877
```

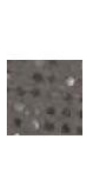
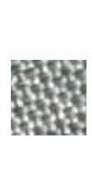
```
Index: 62
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.000946
```

```
Index: 63
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.001126
```



```
Index: 64
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.000636
```

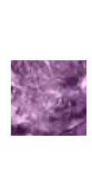
```
Index: 65
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.000636
```

```
Index: 66
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.000355
```

```
Index: 67
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.000099
```

```
Index: 68
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.000099
```

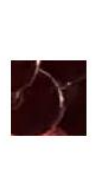


```
Index: 69
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.000153
```

```
Index: 70
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000334
```

```
Index: 71
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000104
```

```
Index: 72
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000108
```



```
Index: 73
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000946
```

```
Index: 74
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.000312
```